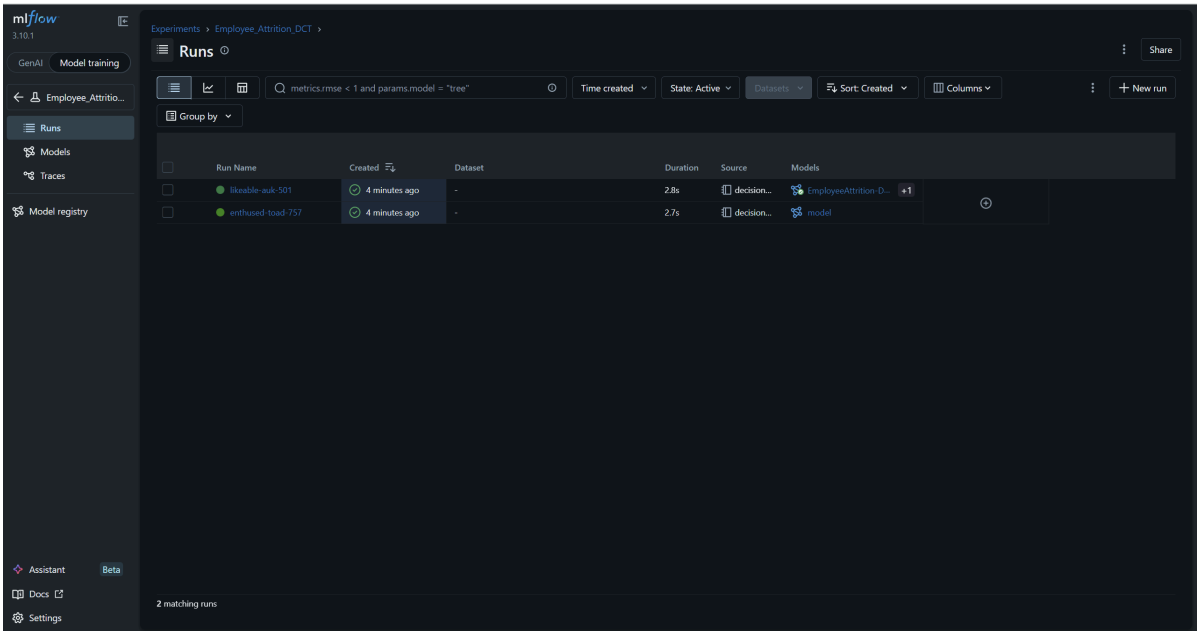
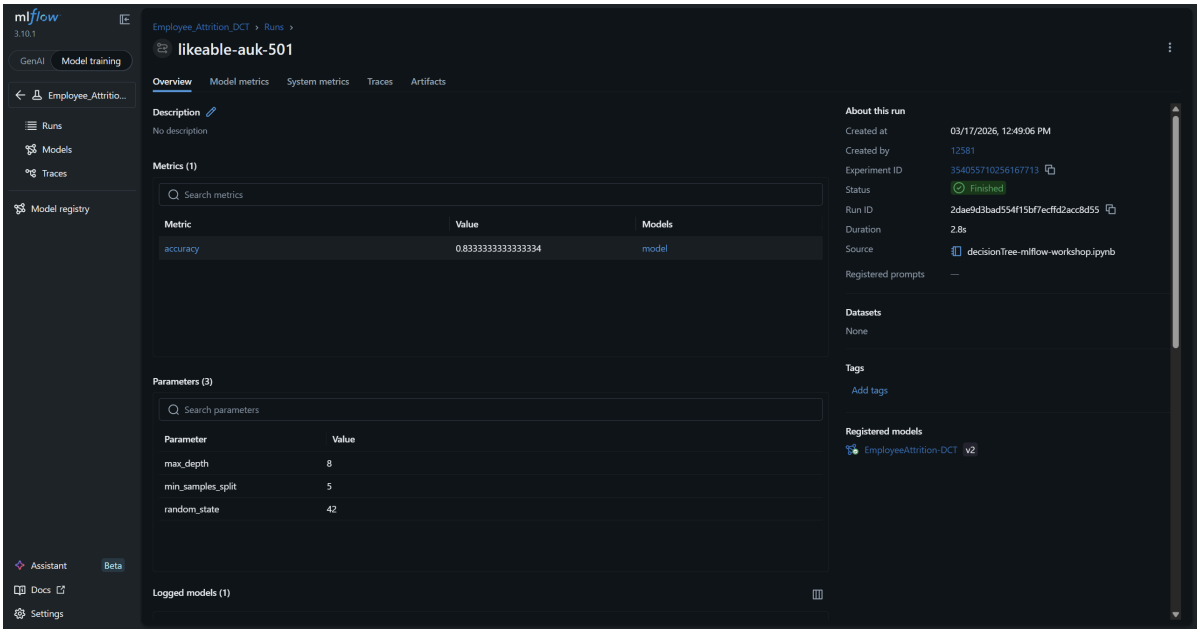
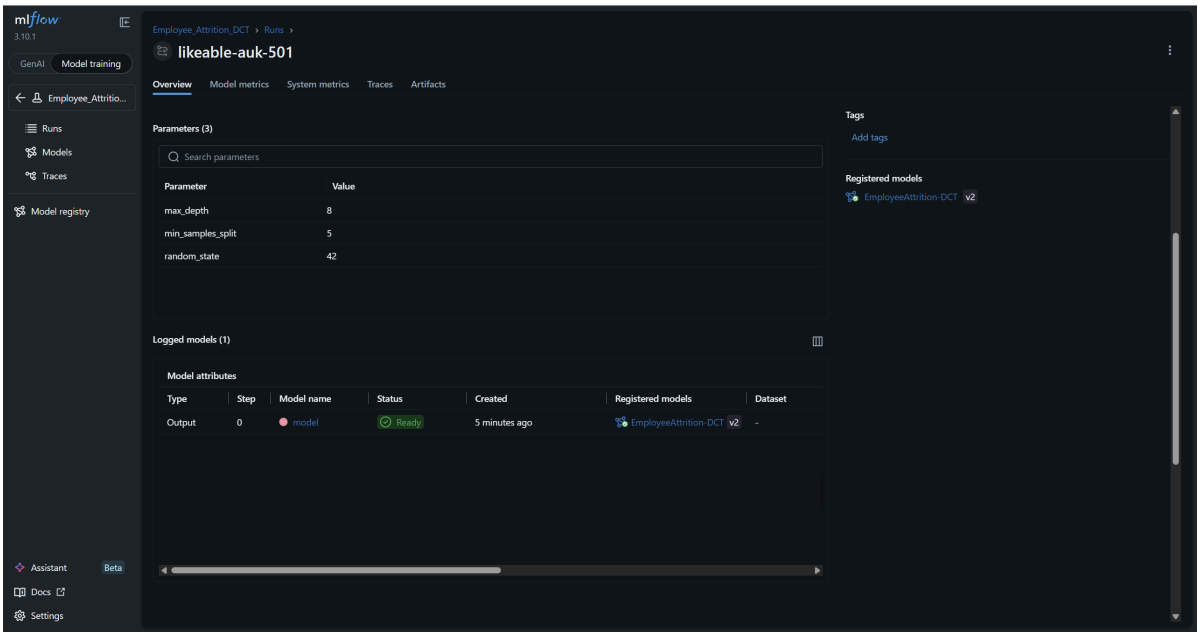
IDA - Individual Workshop

Owner: Sun Sihan








```python
# Log the accuracy metric
mlflow.log_metric("accuracy", accuracy_dt)

# Log the model and register it (same registered model name → version 1, 2, ...)
mlflow.sklearn.log_model(
    dt_classifier,
    artifact_path="model",
    registered_model_name="EmployeeAttrition-DCT",
)

print(f"Run with max_depth={params['max_depth']}, min_samples_split={params['min_samples_split']} -> Accuracy: {accuracy_dt:.4f}")
```

[8] ✓ 5.5s Python

```
c:\Users\12581\AppData\Local\Programs\Python\Python313\Lib\site-packages\mlflow\tracking\_tracking_service\utils.py:184: FutureWarning: The fi
  return FileStore(store_uri, store_uri)
2026/03/17 12:49:03 INFO mlflow.tracking.fluent: Experiment with name 'Employee_Attrition_DCT' does not exist. Creating a new experiment.
2026/03/17 12:49:03 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
2026/03/17 12:49:03 WARNING mlflow.sklearn: Saving scikit-learn models in the pickle or cloudpickle format requires exercising caution because
c:\Users\12581\AppData\Local\Programs\Python\Python313\Lib\site-packages\mlflow\tracking\_model_registry\utils.py:220: FutureWarning: The file
  return FileStore(store_uri)
Successfully registered model 'EmployeeAttrition-DCT'.
Created version '1' of model 'EmployeeAttrition-DCT'.
2026/03/17 12:49:06 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
2026/03/17 12:49:06 WARNING mlflow.sklearn: Saving scikit-learn models in the pickle or cloudpickle format requires exercising caution because
Run with max_depth=4, min_samples_split=2 -> Accuracy: 0.8639
Run with max_depth=8, min_samples_split=5 -> Accuracy: 0.8333
Registered model 'EmployeeAttrition-DCT' already exists. Creating a new version of this model...
Created version '2' of model 'EmployeeAttrition-DCT'.
```

Try to make a prediction with this loaded model

```python
# Load the registered model by name and version (e.g. version 1)
# Use "models:/<model_name>/<version>" or "models:/<model_name>/latest"
loaded_model = mlflow.sklearn.load_model("models:/EmployeeAttrition-DCT/1")

# Make a prediction with the loaded model (use one row from test set)
sample = X_test.head(1)
prediction = loaded_model.predict(sample)
print("Sample features (one row):")
print(sample)
print(f"\nPrediction (Attrition): {prediction}")
```

[9] ✓ 0.0s Python

```
Sample features (one row):
      Age  BusinessTravel  DailyRate  Department  EducationField  Gender  \
1041   28               2        866           2               3       1

      HourlyRate  JobRole  JobSatisfaction  MaritalStatus  MonthlyIncome  \
1041          84        7                0              2           8463

      MonthlyRate  NumCompaniesWorked  OverTime  TotalWorkingYears
1041        23490                   0         0                  6

Prediction (Attrition): ['No']
```